IoT-Enabled Smart bus route monitoring and revenue management system

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lanka's public bus transport sector faces significant challenges in route compliance, operational transparency, and effective use of tracking data. Current GPS-based fleet tracking solutions primarily provide passenger information or basic vehicle monitoring, but lack affordable, event-driven tracking for regulatory enforcement and trip-level verification. Continuous GPS polling increases bandwidth usage and communication costs, limiting scalability for public transport. There is also limited research on event-based IoT tracking tailored to Sri Lankan buses, particularly for detecting route deviations, halt-to-halt movement, device tampering, and offline conditions using approved route data. This research introduces an event-driven IoT processing engine for smart bus route monitoring. On-board devices capture GPS coordinates and timestamps, transmitting data only during significant operational events such as trip start, trip end, halt arrival, halt departure, route deviation, tamper detection, and offline status. The system uses geo-fencing around registered halts and junctions, along with map-matching techniques based on official bus route polylines, to verify route compliance and update trip progress in real time. By transmitting only meaningful events, the system aims to reduce bandwidth consumption while ensuring reliable and auditable transport monitoring. Implementation will use an ESP32-based IoT device, GPS module, Wi-Fi communication, geofencing logic, and backend event-processing services integrated with a regulatory monitoring platform. Expected outcomes include accurate trip tracking, timely off-route detection, reliable halt-to-halt event identification, and improved operational visibility for transport authorities. The solution supports safer, more accountable, and better-regulated public transportation, and offers a cost-efficient monitoring approach suitable for resource-constrained environments.

Keywords— Event-driven IoT, GPS tracking, Smart bus monitoring, Route deviation detection, Geofencing, Map-matching.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
AVL	Automatic Vehicle Location
DSR	Design Science Research
ESP32	Espressif 32-bit Microcontroller
GPS	Global Positioning System
IMU	Inertial Measurement Unit
IoT	Internet of Things
JSON	JavaScript Object Notation
LKR	Sri Lankan Rupee
MCU	Microcontroller Unit
NEO-6M	u-blox NEO-6M GPS Module
NMEA	National Marine Electronics Association
NTC	National Transport Commission
PRPTA	Provincial Road Passenger Transport Authority
SDG	Sustainable Development Goal

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1. INTRODUCTION

1.1 Background and Context

Sri Lanka's public bus transport system is essential for providing affordable daily mobility to workers, students, and the general public. Despite its importance, the sector faces significant operational and regulatory challenges, particularly regarding route compliance, trip verification, and service transparency. A persistent issue is that buses may deviate from legally approved routes, skip designated halts, or operate in ways that are difficult for transport authorities to verify in real time. These challenges negatively impact passengers, regulators, and operators, thereby undermining the effectiveness of transport governance [1], [3].

Global Positioning System (GPS) and Internet of Things (IoT) technologies have demonstrated considerable potential for vehicle tracking, monitoring, and intelligent transport management. IoT-based transport solutions enable real-time data collection, communication, and monitoring across vehicles and infrastructure, making them well-suited for smart mobility applications [1], [3], [4]. GPS-based vehicle tracking systems have also been implemented for real-time monitoring of vehicle movement and operational visibility [2]. However, most existing systems are primarily designed for general fleet tracking, passenger information, or traffic monitoring, rather than for low-cost regulatory route enforcement in public bus operations.

A key limitation in many conventional GPS tracking approaches is the use of continuous polling, where coordinates are transmitted at fixed intervals regardless of whether meaningful operational changes have occurred. Although this provides continuous visibility, it can also produce unnecessary data transmission, higher bandwidth usage, and increased storage overhead. In large-scale public transport deployments, especially in cost-sensitive environments, such overhead can reduce system practicality. Recent transport IoT studies show that efficient event-focused sensing and communication are increasingly important for scalable smart transportation systems [3], [4].

To address this issue, this research proposes an event-driven processing approach in which the on-board IoT device transmits data only when significant transport events occur. These events include trip start, trip end, junction entry or exit, halt detection, route deviation, tamper events, and offline conditions. The reviewed system methodology already defines this event-driven model as the core of the tracking component, supported by geo-fencing and route verification logic . Compared with continuous tracking, this approach is expected to reduce unnecessary communication load while still preserving sufficient evidence for trip monitoring and compliance validation.

Another important concern is off-route movement, where a driver may take an unauthorized road outside the approved route. Existing transport tracking systems may display the vehicle location but often do not provide clear route compliance verification linked to official route geometry or ordered halt and junction sequences. Research in traffic forecasting and intelligent transport systems highlights the importance of accurate route modeling and movement analysis for transport management [5]. In this project, official bus route polylines and geo-fenced halt or junction locations are used together with GPS timestamps to identify halt-to-halt progress and detect route deviations more reliably.

Accordingly, this research focuses on developing a low-cost, event-based IoT tracking engine specifically designed for Sri Lankan public buses. As outlined in the Topic Assessment Form, the novelty of this component lies in its Wi-Fi-based, event-driven design, which transmits only significant operational events while supporting reliable route compliance monitoring under practical deployment constraints. This approach contributes to a more intelligent and accountable transport monitoring mechanism and aligns with broader objectives of sustainable infrastructure, enhanced mobility, and strengthened institutional oversight.

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1.2 Literature Review

Recent progress in intelligent transportation systems shows that GPS, IoT, and route analytics can greatly improve how vehicles are monitored and how much visibility operators have. Al-Fuqaha *et al.* state that IoT allows for large-scale sensing, communication, and coordination between devices in real-world settings, which makes it well-suited for transport applications that need constant awareness [1]. In transportation, GPS tracking has been used for a long time to monitor vehicle movement, estimate locations, and help control fleets. Hannan *et al.* explain that GPS-based tracking supports real-time monitoring, but these systems mostly focus on showing location rather than checking if routes are followed or handling event-based processing [2].

Research in smart transportation has gone further by adding IoT to traffic and transport management. Jain and Jain point out that IoT-based systems can make operations more efficient through connected sensing, monitoring, and automation [3]. Yerroju *et al. also* discuss how IoT-driven traffic management uses real-time data to improve transport intelligence [4]. While these studies show that connected sensing helps with transport monitoring, they mostly look at large-scale traffic or fleet improvements, not at affordable solutions for bus-specific compliance.

Another key area of research is route modeling, travel analysis, and predicting movement. Vlahogianni *et al.* show that intelligent transport systems rely more and more on accurate traffic and route data to help with decisions, predictions, and management [5]. This matters because monitoring route compliance needs both live GPS data and the ability to match those points to the expected route. Still, most tracking systems do not fully combine official route layouts, the order of stops, and event-based data transmission in one workflow focused on enforcement.

Existing commercial systems such as Google Transit, Moovit, and NextBus demonstrate the practical value of real-time bus tracking for passengers, mainly by showing live locations and predicted arrival times. As stated in your Topic Assessment Form, these platforms demonstrate the feasibility of GPS-based public transport tracking, but they are primarily designed for passenger information rather than regulatory monitoring or bus permit enforcement. Government-operated AVL systems used in countries such as Singapore and the UK also monitor route adherence and vehicle performance, but they are usually built within better-funded public transport ecosystems and are not necessarily optimized for low-cost, event-driven deployment in Sri Lankan bus operations .

Academic work has also explored low-cost public transport monitoring frameworks using GPS and wireless communication. Your Topic Assessment Form notes that research on GPS- and Wi-Fi-based PTMS frameworks has examined onboard storage, periodic syncing, route modeling, and deviation detection . These studies provide useful technical foundations, particularly for scalable movement logging and anomaly detection. However, several limitations remain. First, many systems rely on continuous or periodic GPS updates, which create unnecessary communication overhead. Second, most solutions are not specifically designed to detect meaningful public transport operational events such as trip start, halt arrival, halt departure, tamper events, or offline status. Third, they rarely align route monitoring directly with the legal and regulatory requirements of approved bus routes in Sri Lanka.

The reviewed methodology draft for your project introduces an event-based alternative to these limitations. It defines an on-board IoT device that transmits data only when significant operational events occur, such as junction entry or exit, trip start and end, route deviation, halt detection, driver behavior events, and tamper or offline conditions . This is a notable improvement because it reduces data transmission volume while preserving important operational evidence. The same draft also proposes geo-fencing for junction detection, sequence-based map-matching for route compliance verification, and trip segmentation using GPS timestamps and official route polylines . Compared with generic GPS trackers, this creates a more structured and auditable model for bus monitoring.

Despite these advances, the literature and existing systems still leave a clear gap. Current solutions provide either general vehicle tracking, passenger-facing information, or broad IoT transport intelligence, but they do not fully address a low-cost, event-driven, regulatory-oriented monitoring engine designed for Sri Lankan public buses. They also do not sufficiently integrate geofencing, map-matching, halt-to-halt progress detection, route deviation alerts, tamper detection, and offline reporting into a single practical bus monitoring component. Therefore, this research extends the existing body of work by focusing on operationally meaningful events, route compliance verification, and communication efficiency in a real public transport setting.

1.3 Research Gap

Although existing GPS and IoT-based transport systems enable real-time vehicle tracking and monitoring, the majority prioritize passenger information, fleet visibility, or traffic analysis instead of route compliance enforcement [2]–[4]. Many current solutions depend on continuous GPS polling, which increases bandwidth consumption, communication expenses, and storage requirements [1], [3]. Consequently, these systems are less suitable for large-scale, cost-effective deployment in Sri Lankan public bus operations.

A further significant limitation is that existing systems do not effectively integrate official bus route polylines, geofencing, ordered halt or junction sequences, and event-based trip monitoring to detect unauthorized route deviations in a reliable and auditable manner. As noted in the Topic Assessment Form, current systems seldom verify strict adherence to legally approved routes and lack adequate support for regulatory monitoring.

Additionally, many existing tracking solutions do not adequately address critical operational events such as trip start, trip end, halt detection, tamper events, and offline status. There is a clear need for a low-cost, event-driven IoT tracking engine specifically designed for Sri Lankan public buses. This research addresses this gap by proposing a system that transmits only significant operational events and employs geofencing and map-matching to verify route compliance efficiently and accurately.

2. OBJECTIVE

2.1 Main Objective

This research focuses on designing, developing, and validating an event-driven IoT bus monitoring system for Sri Lankan public buses. The system uses GPS and timestamp data from on-board devices to automatically track bus locations and trip progress by detecting key events like trip start, trip end, halts, and route deviations. It also includes map-matching and geofencing to check movement between stops and spot when buses go off their approved routes. By sending data only when important events happen and closely monitoring route compliance, the system aims to help transport authorities improve transparency, cut down on unnecessary data sharing, and strengthen public transport regulation during this project.

2.2 Specific Objectives

2.2.1 GPS Data Acquisition and Device Setup

To configure and deploy an on-board GPS-enabled IoT device capable of capturing accurate bus location coordinates and timestamps during live operations. This objective focuses on establishing reliable data collection from buses and ensuring that the device can continuously detect movement-related information required for trip monitoring.

2.2.2 Event-Based Data Transmission

To design and implement an event-driven data transmission mechanism that sends data packets only when significant operational events occur, such as trip start, trip end, halt detection, route deviation, tamper events, and offline conditions. This objective aims to reduce unnecessary bandwidth usage and improve communication efficiency compared to continuous GPS polling.

2.2.3 Map-Matching and Route Compliance Verification

To implement map-matching techniques using official bus route polylines in order to compare live GPS data with approved routes. This objective focuses on verifying whether buses operate according to their legally assigned routes and identifying off-route behavior when buses travel through unauthorized roads or unexpected route segments.

2.2.4 Geofencing-Based Halt Detection

To develop a geofencing mechanism around registered halts and junctions for detecting halt-to-halt movement events. This objective aims to improve trip progress tracking by identifying when a bus arrives at, departs from, or passes through key operational points along its route.

2.2.5 Device Tamper and Offline Status Monitoring

To design and implement a monitoring mechanism that detects possible device tampering and communication loss. This objective ensures that the system can report suspicious device behaviour or offline conditions in a timely manner, improving reliability and operational transparency.

2.2.6 System Validation through Field Testing

To validate the reliability, accuracy, and practical performance of the proposed event-driven tracking engine through controlled real-world field testing. This objective focuses on evaluating GPS accuracy, event detection correctness, and overall system effectiveness in realistic bus route conditions.

3. METHODOLOGY

3.1 Research Approach

The Design Science Research (DSR) methodology is adopted in this study, as the focus is on designing, developing, and evaluating a novel IT artefact to address a real-world transport monitoring problem. The proposed solution extends beyond theoretical analysis by constructing a functional event-driven IoT-based bus monitoring engine that integrates hardware, communication, and route verification mechanisms.

The use of DSR is appropriate because the research addresses practical challenges in Sri Lankan public bus operations, including route deviation detection, halt-to-halt movement tracking, and inefficient continuous GPS data transmission. This approach enables identification of the problem context, definition of solution objectives, development of the proposed tracking component, and evaluation of its performance through field testing and system validation.

The research follows the main phases of Design Science Research: problem identification, objective definition, artefact design and development, demonstration, evaluation, and communication. Initially, the transport monitoring problem and its operational requirements are examined through existing literature, stakeholder context, and system needs. Subsequently, the event-driven tracking engine is designed and implemented using GPS-enabled IoT hardware, geofencing logic, and map-matching techniques. The developed artefact is demonstrated through controlled route-based testing and evaluated using metrics such as GPS accuracy, route deviation detection performance, halt detection reliability, and communication efficiency.

In summary, DSR offers a suitable and structured research approach for this study by supporting both the creation of an innovative technical solution and the evaluation of its practical effectiveness in a real-world public transport context.

3.2 Technical Methods and Algorithms

The proposed system uses several technical methods to support event-driven bus tracking and route compliance monitoring.

3.2.1 Event-Based GPS Data Acquisition and Transmission

The on-board IoT device captures GPS coordinates and timestamps, but sends data only when important events occur, such as trip start, trip end, halt detection, route deviation, tamper events, and offline conditions. This reduces unnecessary bandwidth usage and communication cost.

3.2.2 Geofencing for Halt and Junction Detection

Geofencing is used to detect when a bus enters or exits registered halts and junctions. This helps the system identify halt-to-halt movement and update trip progress accurately.

3.2.3 Map-Matching for Route Compliance Verification

The system compares live GPS coordinates with official bus route polylines using map-matching. If the bus moves outside the approved route, the system detects it as a route deviation.

3.2.4 Trip Progress Tracking

Trip progress is monitored using sequential GPS events, timestamps, halts, and junctions. This allows the system to track bus movement from one point to another in real time.

3.2.5 Tamper Detection and Offline Reporting

The system also detects device tampering and offline conditions to improve reliability and ensure operational transparency.

3.2.6 Field Testing and Validation

The proposed methods will be validated through field testing to check GPS accuracy, event detection performance, and overall system reliability.

3.3 Tools and Platforms with Justification

Tool / Platform	Type	Purpose	Justification
ESP32	Hardware / Embedded Platform	Acts as the main on-board IoT device in the bus	Chosen because it is low-cost, lightweight, and suitable for handling GPS input and Wi-Fi communication
GPS Module	Hardware / Sensor	Captures real-time bus coordinates and timestamps	Needed to collect accurate location and time data for trip tracking and route monitoring
Wi-Fi Communication	Communication Technology	Transmits significant operational events from the bus to the backend	More cost-effective for event-based transmission and reduces communication cost compared to continuous mobile-data tracking
Backend Processing Framework	Software Platform	Processes incoming GPS events and manages trip progress logic	Required to handle event processing, route monitoring, and system logic efficiently
Database System	Software / Data Storage	Stores event logs, route details, timestamps, and trip records	Needed for structured data storage, retrieval, and monitoring history
Map Services / Route Polyline Data	Mapping / Data Resource	Supports map-matching and route deviation detection	Required to compare live GPS data with approved bus routes and identify off-route movement

Table 1Tools and Platforms with Justification

3.4 Data Collection Methods

Data for this research will be collected from both system-generated and route-related sources. The main data source will be the on-board IoT device, which captures GPS coordinates, timestamps, and operational events during bus movement. These events include trip start, trip end, halt detection, route deviation, tamper events, and offline status.

In addition, official bus route information such as route polylines, halt locations, and junction points will be collected to support geofencing and map-matching. Field testing data will also be gathered by running the device in real transport conditions to observe movement patterns and validate event detection.

The collected data will be used to analyze GPS accuracy, trip progress tracking, route compliance, and overall system reliability.

3.5 Data Analysis Methods

The collected data will undergo analysis to evaluate the performance of the proposed event-driven tracking engine. GPS coordinates and timestamps will be examined to verify trip progress, halt detection, and route movement patterns. Route data will be compared with live GPS data through map-matching to determine whether the bus adhered to the approved route or deviated from it.

Additionally, the detected events will be analyzed to measure system accuracy, communication efficiency, and reliability. Field testing results will be used to assess the system's ability to identify operational events, including trip start, trip end, halt arrival, route deviation, tamper events, and offline status. This analysis will determine the effectiveness of the proposed system for practical bus monitoring and route compliance verification.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Our system uses an event-driven IoT approach to keep bus journeys on track and transparent. Each bus has a GPS device that records key moments—like starting a trip, arriving at a stop, or straying off route—and sends just these important updates to the backend. This smart flow helps transport authorities see real-time progress, check compliance, and receive alerts when something needs attention.

The on-board device watches for moments that matter: trip start and end, arriving and leaving stops, route changes, tampering, or going offline. By sending updates only when these events happen, it keeps communication lean and efficient.

Event updates travel from the bus to the central system over Wi-Fi, where the backend matches GPS data with approved routes and stops. This way, the system quickly checks if a bus is following its path and updates trip status on the go.

All trip details—routes, events, and histories—are safely stored in one place. A user-friendly dashboard lets authorities watch trips in real time, spot issues, and review any incidents quickly.

By bringing together smart event reporting, geofencing, and route checks, this system offers Sri Lanka an affordable, scalable way to keep public buses safe and reliable.

4.1: Main Components of the High-Level Architecture

Component	Function
On-board IoT Device	Captures GPS coordinates and timestamps, detects operational events
GPS Module	Provides real-time bus location data
Wi-Fi Communication Layer	Sends event-based data packets to the backend
Event Processing Engine	Processes incoming events and updates trip progress
Geofencing Module	Detects halt and junction entry/exit events
Map-Matching Module	Compares live GPS data with official route polylines
Database	Stores route data, trip records, event logs, and timestamps
Monitoring Dashboard	Displays trip status, route deviations, and system alerts

Table 2 Main Components of the High-Level Architecture

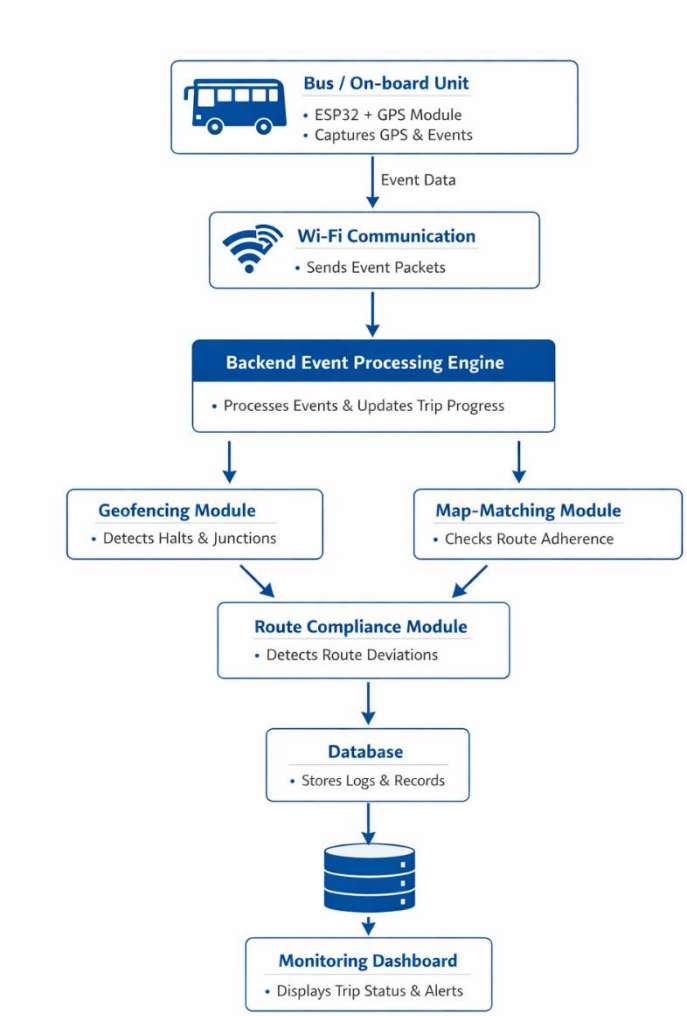


Figure 1 High-Level Architecture

5. USER REQUIREMENTS

5.1 Stakeholders

The bus monitoring system brings together several key groups. Transport authorities rely on it for route checks and transparency. Bus operators and owners gain clearer insights into their fleet's movements. System admins keep everything running smoothly, while inspectors use alerts and data for enforcement. Even passengers, though not direct users, benefit from safer, more reliable public transport thanks to better monitoring.

Stakeholder	Role in the System	Interest / Need
Transport Regulatory Authorities	Main users of the monitoring system	Need route compliance data, trip progress updates, and alerts for route deviation, tampering, and offline status
Bus Operators / Owners	Monitor bus operations and compliance	Need visibility into bus movement, route adherence, and operational performance
Bus Drivers	Indirect users through the on-board device	Need the device to function properly during trips without interrupting operations
System Administrators	Manage and maintain the backend system	Need to maintain route data, event logs, system configuration, and monitoring services
Transport Inspectors / Enforcement Officers	Use reports and alerts for regulatory action	Need reliable evidence of off-route movement and operational violations
Passengers / General Public	Indirect beneficiaries of the system	Need safer, more reliable, and more accountable public transport services

Table 3 Stakeholder Identification

5.2 Functional Requirements

The functional requirements describe the core operations that the proposed bus monitoring system must perform. These requirements focus on GPS data capture, event detection, route compliance verification, and monitoring support.

Requirement ID	Functional Requirement
FR1	The system shall capture GPS coordinates and timestamps from the on-board IoT device.
FR2	The system shall detect significant operational events such as trip start, trip end, halt arrival, halt departure, and route deviation.
FR3	The system shall transmit only event-based data packets instead of continuous GPS streams.
FR4	The system shall compare live GPS coordinates with official bus route polylines using map-matching.
FR5	The system shall detect halt and junction entry or exit using geofencing logic.
FR6	The system shall identify off-route movement when a bus travels outside the approved route.
FR7	The system shall detect and report tamper events and offline device status.
FR8	The system shall store trip records, GPS events, timestamps, and route compliance logs in the database.
FR9	The system shall update trip progress based on detected movement events.
FR10	The system shall display route monitoring information and alerts through a monitoring dashboard.

Table 4 Functional Requirements

5.3 Non-Functional Requirements

The non-functional requirements define the quality attributes the proposed system must satisfy to operate effectively in a real-world public transport environment.

Requirement ID	Non-Functional Requirement
NFR1	The system shall ensure secure transmission and storage of GPS, event, and route monitoring data.
NFR2	The system shall process incoming event data with minimal delay to support timely trip progress updates and route deviation alerts.
NFR3	The system shall operate reliably during live bus operations and report offline or tamper conditions when failures occur.
NFR4	The system shall provide accurate GPS-based event detection and route compliance verification.
NFR5	The system shall be scalable to support multiple buses and routes without major performance reduction.
NFR6	The system shall minimize unnecessary bandwidth usage and communication cost through event-based transmission.
NFR7	The system shall provide a clear and user-friendly monitoring dashboard for transport authorities and administrators.
NFR8	The system shall be maintainable so that route data, geofencing rules, and backend logic can be updated easily.

Table 5 Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed system is the public transport sector in Sri Lanka, especially transport regulatory authorities, private bus operators, and fleet owners. These users need a practical and affordable solution to monitor route compliance, detect unauthorized deviations, and improve operational transparency.

The main customer persona is a transport regulator or fleet manager who is responsible for monitoring multiple buses and ensuring that vehicles operate according to approved routes. This user values low operational cost, reliable event detection, simple dashboard visibility, and evidence-based monitoring for enforcement and decision-making.

6.2 Value Proposition

The proposed system offers a low-cost, event-driven bus monitoring solution that is specifically designed for Sri Lankan public transport operations. Unlike continuous GPS tracking systems, it transmits only significant operational events such as trip start, halt arrival, route deviation, tamper events, and offline status. This reduces bandwidth usage and communication cost while still providing reliable route compliance monitoring. The system therefore delivers both cost efficiency and regulatory usefulness.

6.3 Revenue Model

The system can be commercialized through a combination of:

- Device installation charges for each bus
- Initial system setup fees for operators or authorities
- Monthly or annual subscription fees for access to the monitoring platform
- Maintenance and support services for system updates and technical assistance

This model allows both one-time and recurring revenue streams.

6.4 Cost Estimation

The main costs involved in delivering the solution include:

- IoT hardware costs such as ESP32 devices and GPS modules
- Device installation and testing costs
- Backend development and hosting costs
- Database and dashboard maintenance costs
- Field support and technical service costs

Since the system uses event-based transmission and affordable hardware, the overall operating cost can remain lower than traditional continuous-tracking systems.

6.5 Pricing Strategy

A cost-plus pricing strategy with flexible service packages can be used. A basic package may include the on-board device, core route monitoring, and essential alerts, while advanced packages may include analytics, reporting, and multi-route dashboard features. Pricing should remain affordable for Sri Lankan transport operators while still ensuring the sustainability and scalability of the service.

6.6 Competitive Advantage

The main competitive advantage of the proposed system is its event-driven design, which reduces unnecessary communication and storage costs compared to traditional continuous GPS tracking. In addition, it is specifically tailored for route compliance verification, not just passenger information or live location display. The system also combines geofencing, map-matching, off-route detection, tamper reporting, and offline monitoring in one integrated solution, making it more relevant for regulatory and operational use.

6.7 Intellectual Property Considerations

The intellectual property of the project may include the event-driven processing logic, route compliance verification workflow, system design, and custom software components developed during the research. The source code, backend logic, and device integration design should be documented properly to support ownership and future protection. If the project is further developed for commercial deployment, copyright protection, licensing, or possible patent consideration for unique system mechanisms may be explored.

7. BUDGET AND JUSTIFICATION

The development of the proposed image-based disease detection system requires several resources, including hardware, software tools, cloud services, and human effort. The following budget outlines the estimated costs associated with the research and development process.

DESCRIPTION	AMOUNT (LKR)	TOTAL (LKR)
SOFTWARE		
React.js	6 000	
Spring Boot	10 000	
React native	10 000	
C++	7 000	
PostgreSQL	9 000	
Visual Code	5 000	
Software Cost		47 000
HARDWARE		
Router	6 000	
2 pc	40 000	
IoT Device	9 000	
Total Hardware Cost		55 000
NETWORK		
Data transfer and API usage	8 000	
Hosting and server deployment	7 000	
Total Networking Cost		15 000
MAINTENCE		
Client training and technical support	6 000	
Warranty and support services	10 000	
Security software and monitoring	4 000	
System maintenance agreement	5 000	
Total Maintenance Cost		25 000
Total Project Cost		**142 000**

Table 6 Budget

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) defines the main tasks required to complete the research project and distributes responsibilities among team members. The tasks are organised to ensure balanced workload distribution and alignment with the project timeline.

Task ID	Task Description	Responsible Member	Timeline
T1	Literature review and problem analysis of public bus transport monitoring systems	All Members	Weeks 1-3
T2	Collection of transport operational data, route schedules, and policy information from the Road Passenger Transport Authority	All Member	Weeks 3-5
T3	Development of an IoT-based GPS event monitoring and trip data acquisition module	Member 1	Weeks 5–7
T4	Development of route compliance verification and map-matching algorithm	Member 1	Weeks 7–10
T5	Design and implementation of Policy & Time-Slot Engine for automated scheduling and service classification	Member 2	Weeks 8–11
T6	Development of an incentive and revenue calculation module based on operational distance and service periods	Member 3	Weeks 9–12
T7	Database design and implementation for route data, policy rules, and operational logs using PostgreSQL	All Members	Weeks 10–12
T8	Development of a regulator web dashboard using React for monitoring routes and policy compliance	Members 2 & 3	Weeks 11–13
T9	Development of mobile applications for bus operators to update trips and receive operational notifications	Member 4	Weeks 11–13
T10	Integration of IoT monitoring, policy engine, revenue module, and user interfaces	All Members	Weeks 14–16
T11	System testing and validation using simulated and real operational scenarios	All Members	Weeks 16–18
T12	Performance evaluation, documentation, and final system improvements	All Members	Weeks 18–20

Table 7 WBS

9. GANTT CHART

The Gantt chart illustrates the timeline and sequence of activities planned for the research project. It outlines key phases such as literature review, dataset collection, model development, system implementation, testing, and final documentation. Each task is scheduled across different months to ensure steady progress and timely completion of the project milestones. This structured timeline helps track the progress of development and ensures that the individual components and the overall system are completed within the allocated research period.

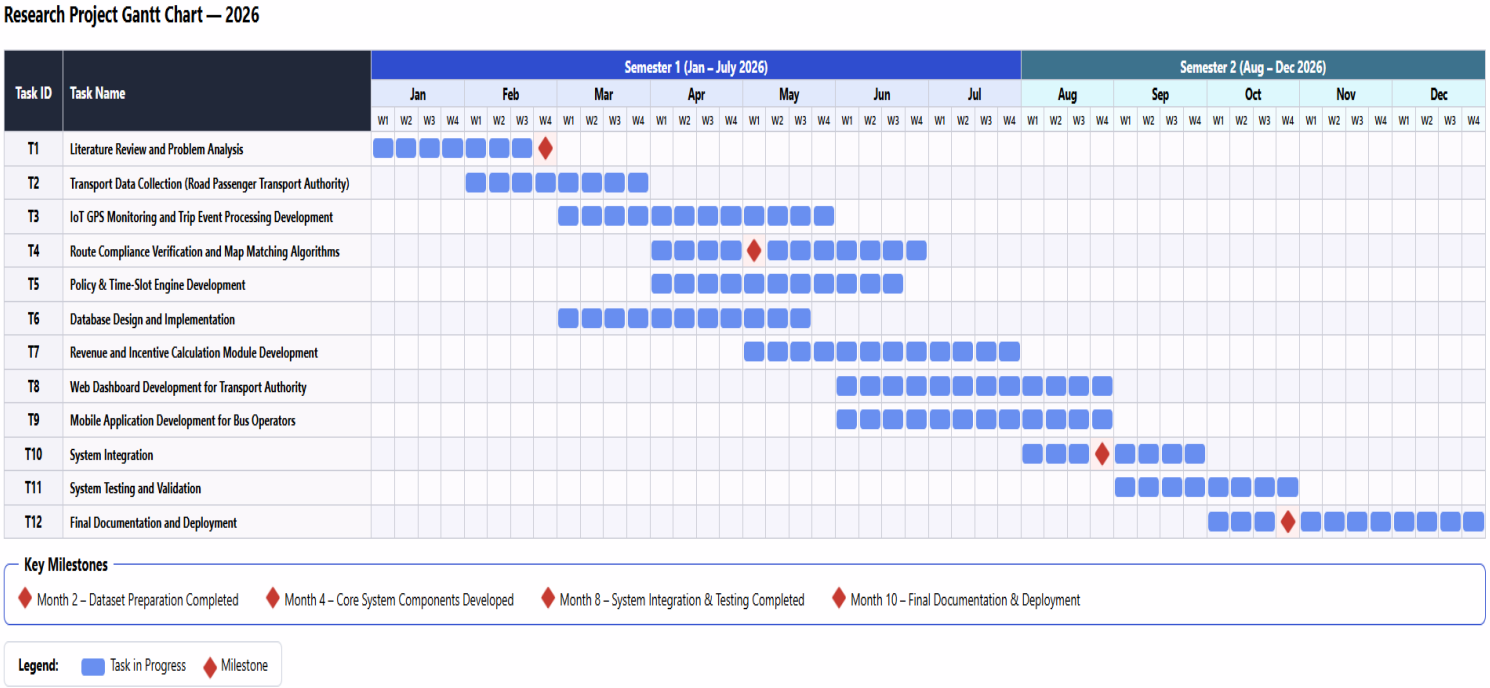


Figure 2 GANTT CHART

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11.APPENDICES

The appendices section includes supporting materials that provide additional technical and research details related to the proposed system. These materials are attached separately to improve clarity and to avoid overloading the main report.

Appendix A – High-Level System Architecture

Includes the overall architecture diagram of the proposed event-driven bus monitoring system, showing the main components such as the on-board IoT device, communication layer, backend processing engine, database, and monitoring dashboard.

Appendix B – Functional and Non-Functional Requirements

Contains the detailed list of functional and non-functional requirements identified for the proposed system, including system behavior, performance, security, usability, and reliability needs.

Appendix C – Budget Estimation

Presents the estimated cost of the prototype, including hardware components, connectivity, field testing, and miscellaneous expenses.

Appendix D – Work Breakdown Structure (WBS)

Includes the task breakdown for the project, showing the major activities and sub-activities planned for design, development, testing, and reporting.

Appendix E – Gantt Chart

Contains the project timeline showing the planned schedule of activities and milestones across the research period.